The Complement Rule Is Excluded Middle: Machine-Checked Probability over Intuitionistic Logic

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Abstract

Kolmogorov's probability is defined within a classical logic framework, whose complement rule becomes equivalent to the law of excluded middle in the limiting case of certainty. Quantum mechanics has shown that this logical substrate is just one of many possible logical modeling choices. We mechanize, in Lean 4 over mathlib, what probability becomes when the event logic is *intuitionistic*: valuations on frames, the algebras of verifiable propositions. Two characterizations anchor the development: (1) the complement rule $v(A^c) = 1 - v(A)$ holds for every valuation exactly when the algebra satisfies excluded middle, and (2) Dempster conditioning agrees with Bayesian conditioning at a piece of evidence exactly when that evidence carries zero slack. On finite frames, every valuation is a classical mass function on points. In the general case, valuation splits into atomic and diffuse parts. Its Boolean core carries a totally monotone Dempster–Shafer belief function. We define slack as the mass that escapes point-valued certainty: it is boundary mass in the measure model. On a semidecidable event, forcing it to zero is exactly as hard as solving the halting problem. We further prove that product valuations are unique on chain products and provably non-unique on the powerset frame, where Fubini fails, show that the naive transcription of Cox's theorem is unsatisfiable, and build the Aczél/Hölder analytic core from scratch.

Keywords: intuitionistic logic, probability, Cox's theorem, belief functions, locales, Lean, mathlib, formalized mathematics

1 Introduction

Classical logic underlies Kolmogorov's probability [Kolmogorov 1933]. The axioms of this probability specify an arithmetic of chance (non-negativity, normalization, σ-additivity). Beneath that arithmetic sits a logic: the events form a Boolean σ-algebra, closed under complement and satisfying excluded middle. The complement rule, $P(A^c) = 1 - P(A)$, is where the two layers touch. It is an arithmetical equation about a logical operation, obeyed by every assignment of degrees of certainty. In the limiting case of all-or-nothing probabilities, it becomes equivalent to the law of excluded middle.

There is no intrinsic reason why probability should be compatible with exactly one logic as different situations are described by different logics. Intuitionistic logic describes computation, verification, and finite-precision observation, where a proposition may be confirmable but not refutable. Linear logic describes resources, where using an assumption consumes it [Girard 1987]. Quantum logic describes measurement, where propositions fail to distribute. If probability is, as the Cox–Jaynes tradition has it, *extended logic* [Cox 1946; Jaynes 2003], the calculus of degrees of certainty over a space of propositions, then the logic of those propositions is a *parameter* of the theory, and Kolmogorov's calculus is the value of that parameter at Boolean logic.

Physics has already introduced a use case for a logic not based in Boolean algebra. Quantum probability lives on orthomodular lattices of projections rather than Boolean algebras [Birkhoff and von Neumann 1936]. Gleason's theorem [Gleason 1957] does for quantum probability what this paper's representation theorems do here: it identifies the states compatible with the event logic. It seems natural to conclude that the Boolean substrate is a modeling choice. In this work, investigate what happens when we change this parameter. Specifically, we ask "what does probability become when the logic of events is *intuitionistic*?".

Intuitionistic logic is the natural first non-Boolean instance to study using the proofs-as-programs paradigm. The choice of (in our case, intuitionistic) logic sits upstream of statistics the way a type system sits upstream of programs: the same syntax of operations, a different set of valid transformations, and some derivations that typecheck classically can either fail or split in two. This logic's algebraic semantics is mature order theory, well supported by mathlib [mathlib Community 2020]: frames, that is, complete Heyting algebras, modelled on the lattices of open sets. The content of the original (classical) probability definitions and results is not lost: complete Boolean algebras are exactly the frames where excluded middle holds, so the classical theory embeds as a special case rather than being replaced.

The choice of logic fixes the event algebra. What remains is to fix the arithmetic on top of it, keeping only the axioms that can be stated without a complement. A *valuation* on a frame Ω is a monotone, normalized, modular map $v : \Omega \longrightarrow [0, \infty]$. For v, modularity means inclusion–exclusion, $v(a) + v(b) = v(a \sqcup b) + v(a \sqcap b)$, and no axiom mentions a complement. In a Heyting algebra the pseudo-complement a^c satisfies $a \sqcap a^c = \bot$ but not, in general, $a \sqcup a^c = \top$. The complement rule therefore degrades to a derived *inequality* $v(a) + v(a^c) \leq 1$, with a gap

$$\mathrm{slack}(a) \;=\; 1 - v(a) - v(a^c)$$

measuring exactly the valuation of the region where excluded middle fails for a. The term *slack*, and this logical reading, are introduced in this paper. The quantity itself is not new: it is the width of the classical Dempster–Shafer belief/plausibility interval [Dempster 1967; Shafer 1976]. The two align on the frame's Boolean core. Dempster–Shafer reads that width as an agent's ignorance. Here it needs no agent: in the concrete model where Ω is the opens of a space carrying a probability measure μ, the slack of an open U is precisely $\mu(\partial U)$, the measure of its topological boundary. The slack of an open set is a structural quantity, not a measure of ignorance. The theory that results recognizably has the same structure as the Dempster–Shafer belief functions, but derived from the underlying logic rather than postulated.

The statistical *operations* are logic-uniform. The conditional is still $v(a \sqcap b)/v(b)$, and the product rule and Bayes symmetry hold verbatim. What varies is the set of *identities* those operations satisfy, and with it the licensed computations. The classical case split over $\{A, A^c\}$, and with it the usual denominator of Bayes' theorem $P(B) = P(B \mid A)\,P(A)+P(B \mid A^c)\,P(A^c)$, degrades to a lower bound whose deficit is a conditional slack. Marginalization is exact only over *genuine* partitions, families that actually join to $\top$. The complement trick $P(A) = 1 - P(A^c)$ degrades to the interval $[\,v(A),\ 1 - v(A^c)\,]$ of width $\mathrm{slack}(A)$, so point statistics becomes interval statistics exactly on the events the logic leaves undecided, and collapses back to points on complemented ones. Updating itself forks: Dempster's rule and Bayesian conditioning, classically indistinguishable, give different posteriors from the same evidence unless its slack vanishes.

The choice of logic changes the query being answered: the value of an event A is $P(\mathrm{int}\,A)$, i.e. the probability of its verifiable part. In the measure model, slack is boundary mass, so a nonatomic distribution queried on intervals behaves entirely classically, since an interval's boundary is two points, which carry no mass. The differences are observable precisely at events with charged boundaries and at semidecidable hypotheses.

A theory this close to excluded middle must be developed with unusual care, because classicality is easy to smuggle in: a single informal case split over $\{A, A^c\}$ silently decides an undecidable event. The risk is documented even classically, as Cox's original theorem required repair after Halpern's counterexample [Cox 1946; Halpern 1999; Horn 2003], and the constructive setting multiplies the opportunities. The direct transcription of the classical Cox statement within an intuitionistic logic setting is not merely unprovable but *unsatisfiable*. The sum rule is irreducible: a disjunction's value is provably not a function of the values of its disjuncts, so modularity must be posited rather than derived. Every appearance of classical logic in this work, including at the meta-theory level, is tracked and justified at the point it is used, as with the computable `Decidable` instance behind the `CoxModel` non-vacuity witness of Section 6.

Full excluded middle is not the only stop between plain intuitionistic logic and Boolean logic. Asking what changes at an intermediate stop is a natural companion question to our main inquiry into intuitionistic probability. Weak excluded middle, $a^{cc} \sqcup a^c = \top$, is one such case. Using our valuation machinery, we can demonstrate that slack collapses to the regularity defect alone under exactly this condition, dropping the independent De Morgan gap.

Contributions. Our contributions are as follows:

1. **The two characterizations** (Sections 3 and 5). The complement rule holds for *every* valuation iff excluded middle holds in the frame, an equivalence isolating the exact content of Van Horn's classical negation axiom R3 [Horn 2003]. Dempster and Bayesian conditioning agree iff the evidence has zero slack;
2. **Structure theorems** (Sections 4 and 5). On finite frames a valuation is a classical mass function on points. In general it splits into atomic and diffuse parts, and on its Boolean core of regular elements it is a belief function;
3. **The Cox program, dissected** (Section 6, Appendix A). The product-rule half is independent of the logic, proved by an Aczél/Hölder generator built from scratch. The sum-rule half (a theorem in classical Cox) admits no such derivation here, as shown by a five-element counterexample, and must therefore be included among the defining axioms of a valuation;
4. **Two grounding models** (Section 7). Any classical probability measure, read on the opens, is a valuation whose slack is boundary mass. On the Sierpiński frame, a slack-free classical readout of halting is uncomputable;
5. **The monad dichotomy** (Section 9). Product valuations are unique on chain products and provably non-unique on the powerset frame, where Fubini fails. The countable fragment composes associatively and commutatively;
6. **Weak excluded middle.** (Sections 2 and 7). The global weak excluded middle condition is exactly extremal disconnectedness, the class of Stonean spaces;
7. **Mechanization**. Everything above is mechanized in Lean 4 [de Moura and Ullrich 2021] using mathlib [mathlib Community 2020], and is (to our knowledge) the first mechanization of non-additive probability over intuitionistic logic.

2 Valuations on Frames

2.1 The Event Logic

Definition 1 (frame)**.** A *frame* is a complete lattice Ω in which finite meets distribute over arbitrary joins,

$$a \sqcap \left(\bigsqcup_{i \in I} b_i \right) = \bigsqcup_{i \in I} (a \sqcap b_i)$$

for every $a \in \Omega$ and every family $\{b_i\}_{i \in I} \subseteq \Omega$. Equivalently, it is a complete Heyting algebra.

Frames are the algebraic semantics of intuitionistic logic and the point-free abstraction of topological spaces. The same objects with morphisms reversed form the category of *locales*, so the distinction is invisible at the level of a single algebra. We say *frame* throughout and reserve *locale* for the spatial and categorical statements (points of a locale, classifying locales) where that word is the standard one. The motivating example throughout is $\Omega = \mathcal{O}(X)$, the opens of a space, with $\sqcap, \sqcup$ as intersection and union and the pseudo-complement a^{c} of an open given by the *interior* of its set complement. In mathlib this is the typeclass `Order.Frame`, and the Heyting operations come with the key asymmetry already in place: $a \sqcap a^{\mathrm{c}} = \bot$ holds in any Heyting algebra (`inf_compl_eq_bot`), while $a \sqcup a^{\mathrm{c}} = \top$ is available only in Boolean algebras. An element with $a \sqcup a^{\mathrm{c}} = \top$ is *complemented*: it satisfies its own instance of excluded middle.

2.2 The Valuation Structure

The central object assigns a degree of certainty to each element of the frame, with inclusion–exclusion as the only interaction axiom:

Definition 2 (`Valuation`)**.** A *valuation* on a frame Ω is a *monotone* function $v : \Omega \to [0, \infty]$ with $v(\bot) = 0$, $v(\top) = 1$ (*normalization*), and, for all $a, b \in \Omega$, *modularity*:

$$v(a) + v(b) = v(a \sqcup b) + v(a \sqcap b).$$

```
structure Valuation (Ω : Type*) [Order.
    Frame Ω] where
  toFun : Ω → ℝ≥0∞
  map_bot' : toFun ⊥ = 0
  map_top' : toFun ⊤ = 1
  mono'    : Monotone toFun
  modular' : ∀ a b, toFun a + toFun b
             = toFun (a ⊔ b) + toFun (a ⊓
                 b)
```

Values live in $[0, \infty]$ (`ENNReal`, extended non-negative reals) following mathlib's measure-theoretic convention, and normalization then confines them to $[0, 1]$ (`Valuation.le_one`). The definition is similar to Kolmogorov's, with the difference being that *no axiom mentions complement*. Classically, this omission is invisible, because the complement rule is derivable from additivity and $a \vee \neg a = \top$. Constructively, this changes the definition of what probability is.

2.3 Slack

Modularity applied to the disjoint pair (a, a^{c}) collapses to $v(a) + v(a^{\mathrm{c}}) = v(a \sqcup a^{\mathrm{c}})$ (`Valuation.add_compl_eq_sup`). So, the join need not be $\top$:

Theorem 3 (`Valuation.add_compl_le_one`)**.** *For every valuation v on a frame Ω and every $a \in \Omega$, $v(a) + v(a^{\mathrm{c}}) \leq 1$.*

This is a *derived* Dempster–Shafer sub-additivity inequality. The deficit

$$\mathrm{slack}(a) = 1 - \big(v(a) + v(a^{\mathrm{c}})\big) = 1 - v(a \sqcup a^{\mathrm{c}})$$

is the valuation of the undecided region. It decomposes canonically. The pair $(a^{\mathrm{cc}}, a^{\mathrm{c}})$, double negation against negation, is always disjoint. However, it is a partition only under *weak* excluded middle ($a^{\mathrm{c}} \sqcup a^{\mathrm{cc}} = \top$). Applying modularity to this disjoint pair splits the slack into two logically independent gaps:

Definition 4 (regular element)**.** An element $a \in \Omega$ is *regular* (or $\neg\neg$-stable) if $a = a^{\mathrm{cc}}$.

Theorem 5 (`Valuation.slack_eq_dnGap_add_deMorganGap`)**.** *For every valuation v on a frame Ω and every $a \in \Omega$, $\mathrm{slack}(a) = \mathrm{dnGap}(a) + \mathrm{deMorganGap}(a)$, where $\mathrm{dnGap}(a) = v(a^{\mathrm{cc}}) - v(a)$ is 0 if a is $\neg\neg$-stable (regular), and $\mathrm{deMorganGap}(a) = 1 - v(a^{\mathrm{cc}} \sqcup a^{\mathrm{c}})$ is 0 if weak excluded middle holds at a.*

The single Dempster–Shafer "ignorance" number is thus resolved into a regularity defect and a De Morgan defect: the first measures failure of double-negation stability, the second corresponds to a named intermediate logic (weak excluded middle). An informal aside on what the first gap measures: constructively, the contrapositive of a universal hypothesis returns only its double-negated form, so for open sets, this specification of probability does not fall victim to Hempel's raven paradox [Hempel 1945]: evidence of the white sock indoors confirms at most a^{cc}, not a itself, and dnGap is exactly the gap between confirming a^{cc} and confirming a itself.

Remark 6 (`Valuation.slack_eq_dnGap_of_isDeMorgan`). The two gaps are logically independent at a single element. However, if weak excluded middle is imposed on the frame for every a ($a^{\mathrm{cc}} \sqcup a^{\mathrm{c}} = \top$, the condition `Order.Frame.IsDeMorgan`), then deMorganGap vanishes identically and $\mathrm{slack}(a) = \mathrm{dnGap}(a)$ for every valuation and every a. This sits strictly between plain intuitionistic logic, where both gaps can be positive, and full excluded middle, where both vanish (`classical_slack_zero`). We do not build on this regime elsewhere, as it is merely a parallel development and an obvious complement to the fully constructive theory this paper focuses on. It easily follows from decomposition for free, and a concrete instance appears in Section 7.

2.4 What Survives without Excluded Middle

Three classical pillars survive intact, and one fails in a precisely delimited way. *Disjoint additivity* is unconditional: $a \sqcap b = \bot$ implies $v(a \sqcup b) = v(a) + v(b)$. *Conditioning* is native: for $v(b) \neq 0$ the posterior $v(a \mid b) = v(a \sqcap b)/v(b)$ is again a valuation (the proof is frame distributivity), and the product rule $v(a \mid b) \cdot v(b) = v(a \sqcap b)$ and Bayes symmetry (and therefore Bayesian updating) $v(a \mid b)\, v(b) = v(b \mid a)\, v(a)$ hold (`Valuation.condVal_mul`, `Valuation.condVal_symm`). *Total probability* holds over any genuine partition, meaning $a \sqcap a' = \bot$ and $a \sqcup a' = \top$: $v(b) = v(a \sqcap b) + v(a' \sqcap b)$. What fails is total probability over $\{a, a^{c}\}$. That family does not tile $\top$, and the conditional masses fall short of $v(b)$ by a conditional slack (`Valuation.cond_add_compl_le`).

The classical fragment. We characterize the elements of Ω on which the classical complement rule holds. The $\neg\neg$-stable (*regular*) elements, which form a Boolean algebra (by Glivenko's theorem), are a natural candidate, but they do not suffice. Modularity constrains the frame join $\sqcup$, whereas the Boolean join of regular elements is the regularized $(a \sqcup b)^{cc}$, for which only the inequality $v(a) + v(b) \leq v((a \sqcup b)^{cc}) + v(a \sqcap b)$ survives (i.e. the 2-monotonicity of Section 5). A regular but uncomplemented element, which is an open ray $(-\infty, 0)$ in the opens of $\mathbb{R}$, under an atomic measure at 0, still carries slack. The correct classical fragment is the *complemented* elements, where $v(a) + v(a^{c}) = 1$ holds exactly (`Valuation.add_compl_eq_one_of_complemented`). On a connected locale that fragment can be as small as $\{\bot, \top\}$. In the Boolean limit, every element is complemented, the slack vanishes identically, and Kolmogorov is recovered (`classical_additivity`, `classical_slack_zero`).

Mixtures. Valuations are closed under finite convex combination (`Valuation.mix`), with every axiom inherited termwise, because modularity is linear. This innocuous fact becomes a characterization in Section 4.

3 Where Classicality Lives

Classicality lives at multiple levels in this work: one meta-theoretic, in the proof assistant itself, and one object-level, in the frame's own logic.

Lean's meta-theory. Lean's foundations include `Classical.choice`, and mathlib is built on it throughout. For example, the order relation on $\mathbb{R}$ and $[0, \infty]$ has no computable `Decidable` instance in general, so any library lemma that case-splits on $x \leq y$ for reals resolves that split via `Classical.propDecidable`. A formalization that never invokes excluded middle at the level of frame elements $a \in \Omega$ can therefore still be classical through the arithmetic it borrows from the library. We track this separately from object-level classicality throughout the paper. Theorem 17 of Section 4 rests only on this ambient baseline and contains no explicit case split. The Scott-continuity recovery theorem (Theorem 16) invokes Lean's `classical` tactic directly, and the `CoxModel` witness of Section 6 goes the other way, replacing the default with an explicit computable `Decidable` instance to keep that one instance genuinely non-classical.

Object-level classicality: the R3 axiom. Van Horn's repaired Cox theorem [Horn 2003] axiomatizes plausibility over classical propositions. Its negation axiom, called *R3*, posits that the plausibility of $\neg A$ is a fixed non-increasing function S of the plausibility of A. Consistency forces the involution $S \circ S = \mathrm{id}$, and therefore the complement rule $p \mapsto 1 - p$. We show that the involutive negation *is* double-negation elimination, so R3 is where classical logic enters every known Cox-style derivation. To do this, we first isolate the exact content of R3's conclusion as a predicate on valuations, then prove equivalence to excluded middle.

```
def Valuation.HasClassicalNegation (v :
    Valuation Ω) : Prop :=
  ∀ a, v a + v aᶜ = 1
```

Theorem 7 (the R3 characterization)**.** *For every frame Ω, every valuation on Ω has classical negation* iff *$a \sqcup a^{c} = \top$ for every $a \in \Omega$.*

The forward direction (`hasClassicalNegation_of_em`) is a calculation. The content is the converse (`em_of_forall_hasClassicalNegation`): wherever excluded middle fails, a valuation with genuine slack must be *manufactured*. Suppose $a \sqcup a^{c} \neq \top$.

Definition 8 (ideal, prime ideal)**.** An *ideal* $J \subseteq \Omega$ is downward closed and closed under finite joins. J is *prime* if $\top \notin J$ and $a \sqcap b \in J$ implies $a \in J$ or $b \in J$.

The prime ideal separation theorem (mathlib's `DistribLattice.prime_ideal_of_disjoint_filter_ideal`) yields a prime ideal J containing $a \sqcup a^{c}$ but not $\top$. The indicator of J's complement,

$$v(x) = \begin{cases} 0 & x \in J \\ 1 & x \notin J, \end{cases}$$

is a valuation. Modularity in the only nontrivial case ($x, y \notin J$) is *exactly* primeness, $x \sqcap y \in J \Rightarrow x \in J \vee y \in J$. And this valuation assigns $v(a) + v(a^{c}) = 0 \neq 1$. So the complement rule is not one convention among many. Quantified over all valuations, it is a statement *about the algebra* that is equivalent to excluded middle.

Sharp valuations are the points of the locale. The construction above is not ad hoc. It is one half of an exact correspondence. Call v *sharp* if it is $\{0, 1\}$-valued, a state of full certainty. Sharpness characterizes exactly the prime-ideal indicators:

Theorem 9 (`sharp_iff_point`)**.** *For every valuation v on a frame Ω, v is sharp iff v is the complement-indicator of a prime ideal.*

This theorem is witnessed by two mutually inverse maps: $v \mapsto Z(v) = \{x \in \Omega : v(x) = 0\}$, which computes the zero-set of a sharp valuation, and the map taking J to its complement-indicator (the valuation of Theorem 7's construction). $Z(v)$ is prime exactly because v is sharp, and the complement-indicator of $Z(v)$ recovers v itself. In locale-theoretic terms, sharp valuations recover the *points* of the locale. A prime ideal is *completely prime* when it is also closed under arbitrary (not just finite) suprema of its members. The finitely prime points of `sharp_iff_point` sharpen to the completely prime points exactly where Scott continuity is added, recovering points of the locale in the sense used by [Coquand and Spitters 2009; Vickers 2008]:

Theorem 10 (`sharp_scott_iff_completelyPrimePoint`)**.** *For every valuation v on a frame Ω, v is sharp and Scott-continuous iff v is the complement-indicator of a completely prime ideal.*

The characterization and the two correspondences together answer the question "where does classicality live": whenever excluded middle fails at some a, the failure is witnessed by a point of the locale.

4 Valuation Representation

Every valuation on a finite frame is a probability mass function on points. On infinite frames, this can fail, and Scott continuity marks the boundary between where it still holds and where it does not.

4.1 Finite Frames Representation

Theorem 11 (Birkhoff duality [Birkhoff 1937])**.** *Every finite distributive lattice is isomorphic to the lattice of lower sets of a finite poset: its sup-irreducible elements form a poset P whose lower sets reconstruct the lattice, and every finite poset arises this way from its own lattice of lower sets.*

Let P be the finite poset of sup-irreducible elements of a finite frame. Then, by Birkhoff duality (`OrderIso.lowerSetSupIrred`), every finite frame is the lattice of *lower sets* of P: a *lower set* $U \subseteq P$ satisfies $p \in U, q \le p \Rightarrow q \in U$, with $\mathrm{Iic}\, p = \{q : q \le p\}$ and $\mathrm{Iio}\, p = \{q : q < p\}$ its principal instances at p. We prove the finite representation theorem below on `LowerSet P`. Transporting it across the isomorphism above then gives the theorem for every finite frame, not just lower sets (`Valuation.exists_pmf_presentation`). Let us define the *mass* of a point as the discrete derivative of v:

```
def Valuation.mass (v : Valuation (
    LowerSet P)) (p : P) :
    ℝ≥0∞ := v (LowerSet.Iic p) - v (
        LowerSet.Iio p)
```

From the mass at every point, we can fully recover v. The masses also sum to 1, forming a probability distribution on P:

Theorem 12 (finite representation, `Valuation.eq_sum_mass`, `Valuation.sum_mass`)**.** *For a finite P, any valuation v on* $\mathrm{Low}(P)$*, and any $U \in$* $\mathrm{Low}(P)$*, $v(U) = \sum_{p \in U} \mathrm{mass}(p)$, and $\sum_p \mathrm{mass}(p) = 1$.*

The proof works by induction on $|U|$: repeatedly remove a maximal point p from U, and use modularity applied to $(U \setminus {\uparrow}p,\ {\downarrow}p)$ to show each removal changes v by exactly $\mathrm{mass}(p)$. `Valuation.toPMF` gives mass the type `PMF` on P, i.e. a classical probability mass function. And since lower sets are exactly the opens of the Alexandrov topology, where interior is the identity, the theorem also reads $v = \mu \circ \mathrm{int}$, i.e. every valuation on a finite frame is the classical measure μ applied after taking the interior. In the finite case, this is the converse of the measure-theoretic bridge of Section 7.

The theorem above represented v as a sum of point masses. The dual view represents v as a mixture of sharp *point-valuations*. For each $p \in P$, let δ_p be the valuation with $\delta_p(U) = 1$ if $p \in U$, and $\delta_p(U) = 0$ otherwise:

Theorem 13 (mixture characterization, `Valuation.eq_mix_deltaPoint`)**.** *On a finite frame, every valuation v satisfies $v = \sum_p \mathrm{mass}(p) \cdot \delta_p$.*

Together with the fact that any mixture of valuations is again a valuation (`Valuation.mix`), this theorem characterizes the valuations on a finite frame exactly: they are the finite convex mixtures of point-valuations, closed under mixing because modularity is linear. Section 6 shows modularity cannot instead be *derived* from more primitive axioms. It must be posited outright, and doing so is exactly what makes every valuation a mixture of sharp, all-or-nothing certainties.

4.2 Infinite Boundary Representation via Scott Continuity

Both halves of the finite theorem fail for infinite frames:

Theorem 14 (obstruction, `exists_valuation_not_point_representable`, `topIndicator_mass`)**.** *On* $\mathrm{Low}(\mathbb{N})$*, the indicator v of $\top$ is a valuation with* $\mathrm{mass}(p) = 0$ *for every point p, and $v(\top) = 1$.*

No point carries any of the mass. The failure is one of approximation: $v(\top)$ is not the supremum of $v(\mathrm{Iic}\, n)$ over the finite stages. In order to recover the representation, we introduce Scott continuity:

Definition 15 (directed set, Scott continuity)**.** A subset $D \subseteq \Omega$ is *directed* if it is nonempty and every pair of its elements has an upper bound in D. A valuation v is *Scott-continuous* if $v(\bigsqcup D) = \sup_{a \in D} v(a)$ for every directed D.

We must require Scott continuity explicitly, as the representation result may fail otherwise. The $\top$-indicator of Theorem 14 witnesses this: it is not Scott-continuous (`topIndicator_not_scott`). Scott continuity is exactly what restores the representation:

Theorem 16 (recovery, `eq_tsum_mass_of_scott`, generalized by `Valuation.isPurelyAtomic_of_scott`)**.** *Let v be a valuation on* $\mathrm{Low}(\mathbb{N})$*. If* $v(\top) = \bigsqcup_n v(\mathrm{Iic}\, n)$ *then* $v(\top) = \sum'_n \mathrm{mass}(n)$.

In general, on any poset all of whose principal downsets are finite, Scott continuity implies zero diffuse part without additional preconditions. However, subtraction in $[0, \infty]$ is truncated at 0, so if $\sum'_p \mathrm{mass}(p)$ were greater than $v(\top)$, the expression $v(\top) - \sum'_p \mathrm{mass}(p)$ would still evaluate to 0. Theorem 17 rules out $\sum'_p \mathrm{mass}(p) > v(\top)$ unconditionally, so this subtraction is never actually truncated and its value is the actual result of the subtraction:

Theorem 17 (general decomposition, `tsum_mass_le`)**.** *For every valuation v on* $\mathrm{Low}(P)$ *and an arbitrary poset P,* $\sum'_p \mathrm{mass}(p) \le v(\top)$.

Rearranging Theorem 17, licensed by the truncation point above, gives $v(\top) - \sum'_p \mathrm{mass}(p) \ge 0$. Every valuation thus splits unconditionally into an *atomic* part carried by points ($\sum'_p \mathrm{mass}(p)$, which is always a sub-probability), and a *diffuse* (nonnegative) remainder $v(\top) - \sum'_p \mathrm{mass}(p)$. This is a constructive Lebesgue-style decomposition. The finite and Scott cases are the purely atomic extremes, and the $\top$-indicator is the purely diffuse one. The proof of Theorem 17 re-runs the maximal-point induction on each finite subset of P, bounding its partial sum by $v(\top)$, then takes the supremum over all finite subsets to bound $\sum'_p \mathrm{mass}(p)$ itself. Its proof does not directly invoke an excluded-middle case split. However, it relies on the (classical) decidability of the ordering of the reals (a default in mathlib). Representing the diffuse part on non-spatial frames remains the open frontier. The obstruction there is again that the frame has *too few points*, the same one as behind the undecidability of Section 7.

5 The Dempster–Shafer Reading

Sections 2–4 independently arrive at the same structure as Dempster–Shafer (DS) theory [Dempster 1967; Shafer 1976], including sub-additive complements, a belief/plausibility interval, and mass functions. This section makes the identification exact in both statics and dynamics. In each case, it locates the precise point where excluded middle makes a difference.

5.1 Total Monotonicity via Constructive Inclusion–Exclusion

The $\neg\neg$-stable (*regular*) elements of a frame form a Boolean algebra with meet $\sqcap$ and join $(a \sqcup b)^{cc}$ (Glivenko). Let us call the resulting algebra the *Booleanization* of this set. A DS *belief function* is a capacity on a Boolean algebra that is n-monotone for every n: $\mathrm{Bel}(\bigvee_i A_i) \ge \sum_{\emptyset \ne T} (-1)^{|T|+1} \mathrm{Bel}(\bigwedge_{i \in T} A_i)$. We show v, restricted to its Booleanization, is such a capacity for every n. The 2-monotone case is immediate from modularity (`two_monotone`). The full tower follows from the following result:

Theorem 18 (inclusion–exclusion, `inclusion_exclusion`)**.** *For every valuation on every frame Ω, every finite family $a : \iota \to \Omega$, and every finite $s \subseteq \iota$, inclusion–exclusion holds with equality for the frame join:*

$$v\Big(\bigsqcup_{i \in s} a_i\Big) = \sum_{\emptyset \ne T \subseteq s} (-1)^{|T|+1}\, v\Big(\bigsqcap_{i \in T} a_i\Big).$$

We state this in Lean additively, with the odd-cardinality half of the signed sum set against the even-cardinality half, so that no $[0, \infty]$ subtraction occurs. The proof is by induction with modularity and frame distributivity.

Replacing the frame join $\bigsqcup_i a_i$ on the equality's other side with its double negation $(\bigsqcup_i a_i)^{cc}$, the Booleanization's own join, yields the following inequality:

Theorem 19 (total monotonicity, `infty_monotone`)**.** *For every valuation on every frame Ω, every finite family $a : \iota \to \Omega$, and every finite $s \subseteq \iota$:*

$$\sum_{\emptyset \ne T \subseteq s} (-1)^{|T|+1} v\Big(\bigsqcap_{i \in T} a_i\Big) \le v\Big(\big(\bigsqcup_{i \in s} a_i\big)^{cc}\Big).$$

The Booleanization's join is given by $(\cdot)^{cc} \circ \sqcup$, and its meets of regular elements are again regular. Interpreted in this way, Theorem 19 states exactly the ∞-monotone tower: v restricted to its Booleanization is a belief function.

The proof of Theorem 19 from Theorem 18 follows from the monotonicity along $a \le a^{cc}$. Here, inclusion–exclusion does *not* fail constructively. It is exact for the intuitionistic disjunction. Applying the double-negation nucleus (a closure operator preserving finite meets) $(\cdot)^{cc}$ to the join on the equality's right-hand side can only enlarge it. This follows from the fact that $a \le a^{cc}$ and v are monotone, which turns the equality into the inequality of Theorem 19. The amount by which the right-hand side is enlarged is exactly the double-negation gap of Section 2.

Additive probability does not become a belief function by weakening an axiom, since modularity for v is never relaxed. What changes is only the choice of join used to state the tower: the frame join $\bigsqcup$ is replaced by its double-negation closure $(\cdot)^{cc} \circ \sqcup$. A belief function is thus the same additive valuation v, evaluated against that closure rather than against the frame join. With $\mathrm{Pl}(a) = 1 - v(a^c)$, the belief–plausibility interval has width exactly the slack (`plausibility_sub_self`). The DS "ignorance" is boundary measure, not psychology.

5.2 The Conditioning Characterization

DS theory carries two update rules that coincide classically. *Geometric* conditioning renormalizes belief, and under the DS dictionary it *is* the localic posterior $v(a \sqcap b)/v(b)$ of Section 2.

Definition 20 (Dempster's rule of conditioning)**.** For a valuation v on a frame Ω and $a, b \in \Omega$, *Dempster's rule of conditioning* [Shafer 1976] states that

$$v(a \,\|\, b) \;=\; \frac{v(a \sqcup b^{\mathrm{c}}) - v(b^{\mathrm{c}})}{1 - v(b^{\mathrm{c}})}.$$

That is, the DS conditioning rule discards the mass incompatible with the evidence and renormalizes by its plausibility. Both the geometric posterior of Section 2 and Dempster's rule of Definition 20 are again valuations (`dempsterCond`). Their modularity is the frame distributivity $(a \sqcup b^{\mathrm{c}}) \sqcap (a' \sqcup b^{\mathrm{c}}) = (a \sqcap a') \sqcup b^{\mathrm{c}}$. Their disagreement is given by the *excluded-middle gap* $\mathrm{emGap}(a, b) = v(a) - v(a \sqcap (b \sqcup b^{\mathrm{c}}))$, i.e. the mass of a outside b's instance of excluded middle:

Lemma 21 (`sup_compl_decomp`)**.** *For every valuation v on every frame Ω and every $a, b \in \Omega$, $v(a \sqcup b^{\mathrm{c}}) = v(b^{\mathrm{c}}) + v(a \sqcap b) + \mathrm{emGap}(a, b)$.*

Dempster's numerator therefore exceeds Bayes's by exactly $\mathrm{emGap}(a, b)$. This gap depends on a, but whether it vanishes at *every* a depends only on the slack at the evidence b:

Theorem 22 (the conditioning characterization, `dempsterCond_eq_condVal_iff_slack`)**.** *For every valuation v on every frame Ω and every $b \in \Omega$ with $v(b) \neq 0$, $\big(v(a \,\|\, b) = v(a \sqcap b)/v(b)$ for every $a \in \Omega\big)$ iff $\mathrm{slack}(b) = 0$.*

The hypothesis $\mathrm{slack}(b) = 0$ eliminates the gap $\mathrm{emGap}(a, b)$ for every a and makes the two normalizers coincide, since $1 - v(b^{\mathrm{c}}) = v(b)$. Conversely, testing at $a := b$, geometric conditioning gives certainty 1, while Dempster gives $v(b)/(1 - v(b^{\mathrm{c}}))$, (which is 1 only if the slack vanishes). Theorem 22 is the valuation-relative analogue of Theorem 7: both are iff statements about the same proposition, $b \sqcup b^{\mathrm{c}} = \top$.

We call the frame-level fact *static* and the valuation-level fact *dynamic* because the former is a fixed structural property of Ω with no notion of updating, while the latter characterizes when a *process* (conditioning v on evidence b) behaves classically. The static characterization is an equation in the frame itself, between $b \sqcup b^{\mathrm{c}}$ and $\top$, with no valuation involved. The dynamic characterization tests the same proposition through v: $\mathrm{slack}(b) = 0$ is exactly $v(b \sqcup b^{\mathrm{c}}) = v(\top)$. Dempster and Bayes therefore coincide at b whenever v cannot distinguish $b \sqcup b^{\mathrm{c}}$ from $\top$, even if the two differ in the frame itself. In the measure model, Dempster and Bayes disagree precisely when the evidence's boundary carries mass.

6 The Cox Program, Dissected

Cox's theorem [Cox 1946; Jaynes 2003] derives the probability calculus from conditions on plausibility. After Halpern's counterexample [Halpern 1999], its repaired forms [Arnborg and Sjödin 2000; Horn 2003] fix the calculus up to monotone regraduation. Every known derivation is classical, and Section 3 showed why: the negation axiom R3 *is* excluded middle. Dropping the R3 axiom has the following effects: product rule is unaffected (as it is logic-free), the sum rule becomes an axiom rather than a result. As a result, the naive constructive statement (strict monotonicity on all of $\mathbb{R}$) is false.

6.1 A Cox Structure without Negation

```
structure CoxModel (Ω : Type*) [Order.
    Frame Ω] where
  pl : Ω → Ω → ℝ                -- pl a b :
      plausibility of a given b
  mono_left : ∀ c, Monotone fun a => pl a
      c
  pl_bot : ∀ c, pl ⊥ c = 0
  pl_top : ∀ c, pl ⊤ c = 1
  F : ℝ → ℝ → ℝ                 -- conjunction
       functional
  conj : ∀ a b c, pl (a ⊓ b) c = F (pl a (
      b ⊓ c)) (pl b c)
  F_assoc : ∀ x y z, F (F x y) z = F x (F
      y z)
  F_strictMono_left : ∀ y, 0 < y →
      StrictMono fun x => F x y
```

The missing R3 axiom can be observed in the omission of any constraint on the relation between $\mathrm{pl}(a^{\mathrm{c}}, b)$ and $\mathrm{pl}(a, b)$. The `CoxModel` restricts monotonicity of F to $0 < y$ to avoid a contradiction with the fact that the boundary axioms force $F(x, 0) = 0$ for all x. `coxModelENNReal` (explicit model on the non-classical chain $[0, \infty]$) satisfies this constraint, as $a = \top$ is provably decidable.

6.2 The Sum Rule Is Irreducible

Classically, additivity is *derived*: R3 turns disjunction into negated conjunction via De Morgan, and the product rule does the rest. Constructively that route is closed, since $a \vee b \neq \neg(\neg a \wedge \neg b)$ in general:

Theorem 23 (`modularity_irreducible`, `no_disjunction_functional`)**.** *Let Ω be the five-element frame of lower sets of the poset $o < a$, $o < b$ with a, b incomparable. Then there is a monotone, normalized assignment q on Ω that is additive on disjoint joins yet not modular, and no binary operation S on values satisfies $q(x \sqcup y) = S(q(x), q(y))$ for all x, y. Moreover, the pairs $(\downarrow a, \downarrow a)$ and $(\downarrow a, \downarrow b)$ have identical value-pairs $(\frac{1}{2}, \frac{1}{2})$ but joins valued $\frac{1}{2}$ and 1.*

Unlike for a conjunction, a disjunction's plausibility cannot generally be expressed as its disjuncts' plausibilities via conditioning. So, a constructive Cox theorem *must* posit the sum-rule half. Modularity, which says that the calculus is closed under mixing certainties, is what posits this, justified by Theorem 13.

6.3 Cox Model Theorem

Strict monotonicity forces $g(-1) < g(0) = 0$, impossible in $[0, \infty]$. For this reason, it is impossible to define a strictly monotone regraduation $g : \mathbb{R} \to [0, \infty]$ with $g(0) = 0$. The repair (strictness only on $[0, 1]$, where plausibilities live) is exactly the same domain restriction that Halpern's critique of the classical theorem turns on. The corrected statement is:

Theorem 24 (`constructive_cox`)**.** *For every frame Ω, every Cox model on Ω whose unconditional plausibility is modular regraduates to a valuation: there exist v : Valuation Ω and g strictly monotone on $[0, 1]$ with $g(0) = 0$, $g(1) = 1$, and for all $a \in \Omega$, $v(a) = g(\mathrm{pl}(a, \top))$.*

The regraduation for the sum-rule half is straightforward (g = `ENNReal.ofReal`). The product-rule half, showing F regraduates to multiplication, on the other hand, requires additional machinery. In particular, we required a proof of Aczél's associativity theorem [Aczél 1966], a statement of pure analysis whose type never mentions Ω. mathlib has no Aczél theorem, no Hölder-style embedding for ordered semigroups on intervals, and no Cauchy functional equation under monotonicity (only under continuity), so we build the analytic core from scratch. A Hölder-style embedding needs two structural facts beyond associativity and monotonicity. The first is *divisibility*: every element must have a 2^n-th root for each n, since the generator is built as the dyadic limit $g(b) = \bigsqcup_n(\text{count of } 2^n\text{-th roots under } b)/2^n$, which cannot even be written down without these roots existing. The second is the *Archimedean property* [Hölder 1901], which rules out elements too fine for any real number to detect, and is what makes an embedding into $\mathbb{R}$ possible at all. Taking these together allowed us to define a normalized, strictly monotone additive generator and hence a strictly monotone multiplicative one on the positive cone (`exists_mul_generator`, `aczelStatement_cone`). Our contribution shortens this hypothesis list on both counts: the Archimedean property is *derived* from continuity and positivity, and the interval form of divisibility needed by the dyadic construction is itself a consequence of the intermediate value theorem (`exists_diag_eq`) applied to `Scale`'s weaker, plain divisibility hypothesis. The constructive/classical divide is therefore *entirely* in the sum rule: this whole apparatus is a statement of pure analysis whose type never mentions Ω. Appendix A gives the construction. The residual analytic gap is discussed in Section 9.

7 Two Grounding Models

Two concrete models demonstrate that our abstract theory is non-degenerate. One shows every classical measure *contains* an intuitionistic probability. The other shows the intuitionistic theory cannot collapse back without solving the halting problem.

7.1 Measures through the Interior Operator

Gödel–McKinsey–Tarski interpret intuitionistic logic in modal S4: propositions are *open* sets, and the modal necessity operator $\Box$ of S4 is interpreted as the topological interior, written $\mathrm{int}(\cdot)$ below [Gödel 1933; McKinsey and Tarski 1944]. Our valuations quantify that same translation:

Theorem 25 (`Measure.toValuationOpens`)**.** *Let μ be a probability measure on a space X, $A \subseteq X$, and define $v(A) = \mu(\mathrm{int}\, A)$. Since $\mathrm{int}\, U = U$ for open U, v restricts on the frame $\mathcal{O}(X)$ to $U \mapsto \mu(U)$, and this restriction is a valuation.*

Modularity is $\mu(U \cup V) + \mu(U \cap V) = \mu(U) + \mu(V)$. The slack of an open U is then $1 - \mu(U) - \mu(\mathrm{int}\, U^{\complement})$, which is the measure of the boundary ∂U. We can now prove that generally, slack is exactly the boundary mass:

Theorem 26.
(`toValuationOpens_slack_eq_frontier`)
For any probability measure μ on a topological space X and any open U, $\mathrm{slack}(U) = \mu(\partial U)$.

The Heyting complement of U in $\mathcal{O}(X)$ is the interior of its set complement. So, the undecided region, the complement of $U \sqcup U^{c}$, is exactly $\overline{U} \setminus U = \partial U$. The general sub-additivity inequality of Section 2 still holds here (`interiorMeasure_add_compl_le`). This identification of slack with boundary mass also exhibits the modal identification $\mathrm{Bel} = P(\Box\,\cdot)$ of the DS literature [Ruspini 1986] arising from the logic itself. Theorem 12 is the converse in the finite case (in the general case, the converse is the open problem, see Section 4).

To characterize weak excluded middle spaces, we give the (standard) definitions required:

Definition 27 (extremally disconnected)**.** A topological space X is *extremally disconnected* if the closure of every open set is again open.

Definition 28 (Stone space, Stonean space)**.** The *Stone space* of a Boolean algebra B is the space of ultrafilters on B (equivalently, the 2-valued homomorphisms $B \longrightarrow \{0, 1\}$), topologized by the clopen basis $\{U : b \in U\}$ for $b \in B$. *Stone duality* [Johnstone 1982] identifies Boolean algebras with exactly the compact, Hausdorff, totally disconnected spaces arising this way. A *Stonean space* is a compact Hausdorff space that is also extremally disconnected. Equivalently, it is the Stone space of a *complete* Boolean algebra.

Remark 29 (`isDeMorgan_opens_iff_extremallyDisconnected`). Section 2 noted that imposing weak excluded middle on the whole frame collapses slack to pure double-negation instability. For $\Omega = \mathcal{O}(X)$, this global condition holds exactly when X is extremally disconnected.

An immediate corollary gives the characterization:

Corollary 30 (`isDeMorgan_opens_iff_stonean`)**.** *Given a compact Hausdorff space X, weak excluded middle on $\mathcal{O}(X)$ holds exactly when X is a Stonean space.*

7.2 The Computability Guard

The usefulness of the work done in this paper relies on the non-degeneracy of the spaces we study (i.e. ones where slack $\equiv 0$ does not always hold). We must provide a witness for this non-degeneracy, and we do so using computability and decidability concepts:

Definition 31 (semidecidable proposition)**.** A proposition is *semidecidable* if it is an open in the Sierpiński topology on `Prop` [Escardó 2004; Rosolini 1986].

On the frame $\mathcal{O}(\mathbb{S}) = \{\bot, \mathit{halts}, \top\}$ (containing three open sets), the Σ_1 asymmetry is a theorem:

Theorem 32 (`haltsOpen_compl`)**.** $\mathit{halts}^c = \bot$.

Every nonempty open of $\mathbb{S}$ contains True, so $\bot$ is the only open disjoint from *halts*. Thus, no nonempty open of $\mathbb{S}$ is evidence that the computation does not halt.

We now instantiate the `Measure.toValuationOpens` construction of Section 7.1 at $X = \mathbb{S}$, allowing us to compute the slack:

Theorem 33 (`slack_eq_boundary`)**.** *Let $\mu = p \cdot \delta_{\mathsf{True}} + (1 - p) \cdot \delta_{\mathsf{False}}$, and let v be the resulting valuation. Then*

$$\mathrm{slack}(\mathit{halts}) \;=\; 1 - p \;=\; \mu(\partial\{q \mid q\}) \;=\; \mu\{\mathsf{False}\}.$$

Since *halts* is already open, $v(\mathit{halts}) = \mu(\mathit{halts}) = p$, which we can use to prove the above theorem. The measure of the slack (i.e. the measure of the outcome when the computation never halts) is the same as the measure of single point `False` in a two-point space. The DS "ignorance" of Section 5 is, in this model, exactly this mass: the probability on the one outcome that no finite observation can ever resolve. This probability is a fact about the model, not about anyone's ignorance.

Definition 34 (halting problem)**.** For a code c and input n, the *halting problem* is the predicate $H(c, n) =$ "the machine with code c halts on input n," (`ComputablePred`).

Slack cannot be eliminated by simply declaring a classical answer for each code. The family of valuations defined below attempts to do this:

Definition 35 (sharp classical readout)**.** For a given n, the *sharp classical readout* is the family of slack-free valuations $\rho_n(c) \in \mathrm{Valuation}(\mathcal{O}(\mathbb{S}))$, with $\rho_n(c)(\mathit{halts}) = 1$ if $H(c, n)$ holds, and $\rho_n(c)(\mathit{halts}) = 0$ otherwise.

However, this family is provably not computable:

Theorem 36 (`sharpReadout_not_computable`)**.** *For every n, the predicate $c \mapsto (\rho_n(c)(\mathit{halts}) = 1)$ is exactly $H(\cdot, n)$, hence not computable (*`ComputablePred.halting_problem`*).*

We can conclude that slack is not a defect the constructive theory: any axiom strong enough to force classical collapse demands of its models to compute the uncomputable. For the sum rule, this guard plays the role the explicit `CoxModel` instance plays for the product rule (Section 6): each axiom set in the paper comes with non-degenerate models satisfying it.

8 Related Work

Informal precedents. Pairing probability with intuitionistic logic is not a new idea, though the literature on it is scattered. Weatherson develops intuitionistic probability axiomatically and defends it with Dutch-book arguments [Weatherson 2003]. Paris treats non-classical uncertain reasoning broadly [Paris 1994]. The identification $\mathrm{Bel} = P(\Box\,\cdot)$ connecting belief functions [Dempster 1967; Shafer 1976] to modal logic is classical [Ruspini 1986]. On the localic side, valuations on frames are standard objects of constructive measure theory, with Vickers's work and the Coquand–Spitters development as canonical references [Coquand and Spitters 2009; Vickers 2008]. These normally build in Scott continuity and target additive, Riesz-flavored goals. Section 4 shows Scott continuity to be a substantive assumption (it decides point-representability on the frames we examine), so our base structure omits it. The Cox line, through Cox, Jaynes, Halpern's counterexample, and the Van Horn and Arnborg–Sjödin repairs [Arnborg and Sjödin 2000; Cox 1946; Halpern 1999; Horn 2003; Jaynes 2003], is classical throughout. Van Horn observes that dropping axiom R3 yields a two-number theory, exactly the belief/plausibility pair our slack separates. To our knowledge, no prior work, informal or formal, states either characterization theorem: complement rule $\iff$ excluded middle (Theorem 7), or Dempster = Bayes $\iff$ zero slack at the evidence (Theorem 22).

Prior formalizations. Three bodies of mechanized work are adjacent, and none overlaps. ALEA [Audebaud and Paulin-Mohring 2009] formalizes a discrete distribution monad in Coq for randomized programs: additive probability, no event logic in our sense. Recent work in cubical Agda formalizes synthetic probability via Markov categories [Sargsyan et al. 2023], where again the objects are additive distributions. Closest is the HoTT/Coq line of Faissole and Spitters on *valuations* over lower reals [Faissole and Spitters 2017], done constructively with Scott continuity and aimed at integration (Riesz) and a Giry-style valuation *monad*. Their valuations are the additive, continuous cousins of ours, and their monad is the Scott-continuous instance of the question examined in Section 9. The delta of this paper is not the word "valuation" but the development on top of it: the non-additive complement regime, slack and its decomposition, both characterization theorems, the DS bridge with total monotonicity and the two conditionings, the representation/mixture/Scott-atomic suite, the corrected Cox theorem with its irreducibility counterexamples, the Aczél construction, product rigidity, and the computability guard. We claim the first mechanization of that development, not of valuations on locales as such.

Non-Boolean probability elsewhere. Quantum probability on orthomodular lattices, with Gleason's theorem [Birkhoff and von Neumann 1936; Gleason 1957], is the long-standing precedent for re-basing probability on a non-Boolean event logic. Free probability, fuzzy measures, and imprecise probability [Walley 1991] vary other parameters (independence, additivity, single-valuedness). Wolpert and Kinney model mathematics itself as a stochastic process, replacing binary theoremhood with a probability distribution [Wolpert and Kinney 2024]. Theoremhood is a semidecidable event, so the computability guard of Section 7 bears directly on that program: slack-free credences about such events are uncomputable. Our variation is specifically the *logic*, holding the Cox-style methodology fixed. Within DS theory, the two conditionings and their inequivalence are classical [Fagin and Halpern 1991; Shafer 1976]. What is new in Theorem 22 is the exact logical characterization of their agreement, and its mechanization.

9 Conclusion: Probability with the Logic as a Parameter

We set out from the observation that Kolmogorov's axioms [Kolmogorov 1933] fix a logic before they fix an arithmetic, and that nothing in the concept of graded belief privileges the Boolean choice. The machine-checked results confirm this at a precision informal treatments do not reach. The entire classical surplus is one axiom, whose statical face is the complement rule (Theorem 7) and whose dynamical face is the agreement of Dempster and Bayesian conditioning (Theorem 22). Everything else survives the intuitionistic substrate intact: conditioning, Bayes symmetry, disjoint additivity, total probability over genuine partitions, and inclusion–exclusion itself (Theorem 18). What the weaker logic adds is structure, not damage: slack with its two-gap decomposition, the atomic/diffuse divide, the belief-function reading through $(\cdot)^{cc}$, and a computability-theoretic reason the additions are not optional.

Open problems. Three are mathematical. First, representing the diffuse part on non-spatial frames (the localic Riesz direction). Second, closing the analytic residue of the Aczél development, namely continuity of the generator and the off-cone extension (the cone results and the log-sum-exp archetype are done). Third, a genuine uniqueness theorem: deriving modularity from a more primitive desideratum, knowing (Theorem 23) that it cannot come from disjunction data alone, with the mixture characterization (Theorem 13) as the current best justification. A fourth is foundational hygiene at scale: a machine-checked Dutch-book theorem in the style of [Weatherson 2003], which to our knowledge no proof assistant has for *any* logic.

Functoriality and the monad question. Within a fixed doctrine, that is, a fixed choice of logic, the connection to the categorical picture is concrete. Pushforward along a frame homomorphism $f : \Omega \to \Omega'$ sends a valuation v on Ω' to $(f_* v)(a) = v(f(a))$ on Ω, and this preserves all four valuation axioms, making $\mathrm{Val}(-)$ a functor on locales, the localic face of the functor part of the Giry monad [Giry 1982]. This is mechanized end to end (`Valuation.comap` builds the pushforward, and `comap_id`/`comap_comp` check the two functor laws against identity and composition of frame homomorphisms). The point valuations δ_p of Section 4 are the unit, already formalized, and Theorem 12 shows that on finite frames Val agrees, on objects, with the classical finite-distribution monad on points. The multiplication is where the substance is. Flattening a valuation of valuations is integration, hence product valuations and Fubini, and by Kock's identification of commutative monads with Fubini theorems [Kock 2012], whether statistics-as-kernel-composition transports to a weaker logic is precisely the question whether that logic's valuation functor is a commutative monad. Two machine-checked results delimit the multiplication from both sides.

On the chain product, the question of *which* product valuation to take does not arise, since the frame $\mathrm{Low}(\mathbb{N} \times \mathbb{N})$ is *rigid*: every lower set is a *finite* union of rectangles, because beyond a stabilization height every occupied column is full, so the full columns and finitely many rows account for everything (`exists_rect_decomposition`). The rigidity mechanism is general. By the inclusion–exclusion equality, two valuations agreeing on any $\sqcap$-closed family agree on all finite joins of its members (`eq_on_finsetSup_of_eq_on`), and the chain product is the instance where that family is the rectangles. The product valuation there, whenever it exists, is therefore unique, with no continuity assumed (`Valuation.product_unique`).

On the powerset frame $\mathrm{Set}(\mathbb{N} \times \mathbb{N})$, whose elements are far from finite unions of rectangles, non-uniqueness is a theorem. The indicator of an ultrafilter is a valuation (`ultrafilterValuation`), and any ultrafilter on a product splits rectangles through its marginal pushforwards (`rect_mem_iff`). Taking a free ultrafilter $\mathcal{U}$ on $\mathbb{N}$, the two iteration orders of its tensor with itself and its diagonal pushforward carry identical values on *every* rectangle, yet they differ on the upper triangle $\{(m, m') \mid m < m'\}$ (`product_valuation_not_unique`). The two orders of iterated integration of the same marginal data therefore disagree: Fubini fails, concretely (`fubini_fails_for_valuations`). Compare the Fubini pathologies of non-additive integration [Denneberg 1994; Ghirardato 1997]. By Kock's correspondence, no canonical commutative multiplication exists for unrestricted valuations. On the powerset frame these valuations are finitely additive probability charges in the sense of [Rao and Rao 1983], so this half of the dichotomy is not specific to intuitionism: the freedom is already present in the classical finitely additive setting, and dropping continuity is what admits it. A monad therefore requires restricting either the frames or the valuations.

The obstruction to a monad. The question upstream of the multiplication is worth stating precisely, though what follows is an assessment rather than a theorem. A monad needs a category for Val to be an endofunctor on. The space of valuations must itself be an object, a "space" whose opens one can speak of. For Scott-continuous valuations that space is a locale and monad structure is available, constructively, through the probabilistic powerdomain line [Coquand and Spitters 2009; Jones and Plotkin 1989; Vickers 2008]. Even there, exhibiting a cartesian closed category closed under the construction is the Jung–Tix problem, open since 1998 for the classical probabilistic powerdomain functor itself [Jung and Tix 1998], though reframed versions of the category have recently been settled. Dropping continuity makes the difficulty prior rather than merely worse, since the standard presentations of the valuation space use continuity essentially.

One candidate route to a well-posed question does exist. The localic treatments present the space of continuous valuations as the classifying locale of a propositional geometric theory [Coquand and Spitters 2009; Vickers 2008], with generators asserting $q < v(a)$ for rational q and axioms for monotonicity, normalization, modularity (expressible with rational witnesses and countable disjunctions), and continuity. Deleting the continuity axiom leaves a geometric theory, so a classifying locale still exists on general grounds, and classically its points are exactly the valuations of this paper. We have verified none of the structure this candidate would need. The unit looks unproblematic, but the multiplication requires integrating the evaluation maps against a merely modular, discontinuous valuation, where the sum of two lower-real-valued maps is an infinite join, exactly where the diffuse part of Theorem 14 leaks. We offer the formulation as a target, not a result.

A more modest strategy, with a better track record, is to carve out a fragment rather than construct the whole monad. On the measure-theoretic side, the failure of cartesian closure for measurable spaces was answered by quasi-Borel spaces [Heunen et al. 2017]. On the domain-theoretic side, commutativity of the full valuations monad on DCPO is open, and the response is a hierarchy of commutative *submonads* generated from simple valuations [Jia et al. 2021]. Our decomposition theorem names the corresponding fragment here: the atomic part. As a first machine-checked step in that direction, we show that the countably-presented fragment composes correctly with no space structure needed at all. A countable mixture of valuations is a valuation (`mixCountable`), a mixture of mixtures flattens to a mixture with product weights, which is discrete Chapman–Kolmogorov (`mixCountable_mixCountable`), the one-point mixture is the identity (`mixCountable_unique`), and iterated countable mixing is order-independent (`mixCountable_comm`), all by the unconditional Fubini of $[0, \infty]$ sums over sigma types. This is associative, unital, commutative kernel composition for countably-atomic data, in direct contrast with the non-uniqueness above. It is not yet a monad: the presentations are not canonical, we do not quotient by "presents the same valuation," and the diffuse part is exactly what escapes every countable presentation.

In summary, the unit, the finite level, and countable kernel composition with its commutativity are proved. Rigidity on chain products and non-uniqueness on the powerset frame are proved, so the general multiplication is settled: it is not canonically determined. What remains is the ambient category, a formulation problem before it is anything else.

Future work on the monad. Two continuations are concrete. First, quotient the countable presentations by "presents the same valuation" and package the result as a Kleisli category of countably-atomic kernels, checking that mixing descends to the quotient. That would upgrade composition-on-data to an honest categorical object. Second, mechanize the classifying-locale formulation sketched above, which requires infrastructure mathlib currently lacks entirely: classifying locales of propositional geometric theories. Both continuations inherit the boundary drawn here: whatever structure exists must live on a fragment, because the unrestricted multiplication is now provably not canonical.

Across doctrines, by contrast, we make no functoriality claim. There is no ambient category in which "logic $\mapsto$ its probability theory" is a functor, and each column requires rederiving what a state *is* from the desiderata. What is uniform across columns is the method, and this paper's execution of it is the Heyting column.

Future work beyond intuitionistic logic. Linear logic is a natural next column: its event algebras are quantales, its conjunction is resource-sensitive, and a "valuation" there should grade the *consumption* of evidence. The analogue of Gleason's theorem or of our representation theorems, that is, which states are compatible with the algebra, is open, as is the right analogue of slack for the failure of contraction. Existing probabilistic semantics of linear logic, probabilistic coherence spaces [Danos and Ehrhard 2011], start from the additive side rather than from the logic, so a Cox-style derivation over a quantale remains to be done. We expect the method used here, state the classical desiderata, delete the axioms the logic no longer licenses, and let a proof assistant referee what survives, what splits, and what becomes unsatisfiable, to transfer directly.

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A The Aczél Development

This appendix sketches the from-scratch construction behind the product-rule half of Theorem 24. The main development does not depend on it. The target shape is Aczél's associativity theorem [Aczél 1966]: a suitably regular associative $F : \mathbb{R} \to \mathbb{R} \to \mathbb{R}$ regraduates to multiplication, $G(F\,x\,y) = G(x) \cdot G(y)$ for all x, y and a strictly monotone G. mathlib offers no route. It has no associativity representation, no Hölder-style embedding of ordered semigroups on intervals, and no Cauchy functional equation under monotonicity (only under continuity).

Hypotheses. A `Scale` bundles: associativity of F, *divisibility* $\forall c\, \exists x,\ F\,x\,x = c$, positivity on the cone $x < F\,x\,c$, joint order-preservation, joint continuity, a unit u, and identity at $-\infty$, meaning $F\,x\,z \to x$ as $z \to -\infty$. Order-preservation was absent from our first bundle, and the additivity proof needs it. Divisibility is taken as a hypothesis in the bundle, although its interval form is a consequence of the intermediate value theorem (`exists_diag_eq`). Non-vacuity: *log-sum-exp*, $F\,x\,y = \log(e^x + e^y)$ with $u = 0$, satisfies every field (`logSumExpScale`), and for it the intended generator is globally exp.

Construction. Write $a^{(n)}$ for n right-associated copies of a under F (`Fpow`). Associativity alone makes $n \mapsto a^{(n)}$ a semigroup morphism from $(\mathbb{N}, +)$, with no commutativity needed. Commutativity is a *corollary* of the generator, not a prerequisite. One fact usually assumed is here *derived*: unboundedness of the iterates. Were they bounded, monotone convergence would give a fixed point $F\,L\,a = L$, contradicting positivity, and unboundedness yields the Archimedean property. Divisibility supplies a tower of 2^n-th roots $u \succ u^{1/2} \succ u^{1/4} \succ \cdots$ (`droot`) with the coherence law that 2^ℓ fine copies recompose one coarse mark. Counting how many copies of the n-th root fit under b (`dcount`, well defined by the Archimedean property) gives dyadic approximants $g_n(b) = \mathrm{dcount}(b, n)/2^n$. The refinement sandwich $2N + 1 \le N' \le 2N + 2$, where N and N' are the counts at consecutive refinement levels n and $n+1$, makes them monotone and Cauchy, and the generator is their limit, $g(b) = \bigsqcup_n g_n(b)$.

Results (all `sorry`-free, standard axioms). On the cone $\{x \mid u \le x\}$ we prove three things. Normalization: $g(u) = 1$. Additivity: $g(F\,x\,y) = g(x) + g(y)$, by squeezing super- and sub-additive counting bounds ($\mathrm{dcount}\,x + \mathrm{dcount}\,y + 1 \le \mathrm{dcount}(F\,x\,y) \le \mathrm{dcount}\,x + \mathrm{dcount}\,y + 3$) against a vanishing 2^{-n} error term. And *strict* monotonicity, proved without any continuity of g by a count-amplification argument: a shrinking root inserted between $x < y$ contributes a full extra block of 2^{n-M} copies, M being the refinement level of the inserted root. Exponentiating gives the multiplicative form, a strictly monotone G with $G(F\,x\,y) = G(x)\,G(y)$ on the cone (`exists_mul_generator`), which is the conclusion shape of Aczél's theorem restricted to the cone (`aczelStatement_cone`). A bounded reorientation $\bar{G} = \exp(-g)$ transports it to the $[0, 1]$-valued conjunctive orientation of Section 6.

Remaining gap. The constructed g is discontinuous at the unit (it is 0 below u), so matching the classical statement verbatim still needs the off-cone extension (a group completion) and continuity of the extension. That is genuine analysis, open here, but *witnessed*: for the log-sum-exp archetype the global continuous generator exists (`hasOrderedGenerator_logSumExp`). Of Hölder's classical hypotheses, the Archimedean property is a theorem in this development rather than an assumption, and the interval form of divisibility follows from the intermediate value theorem. Only the embedding's global regularity remains.